CHAPTER 10

Trigonometry: circle, radians and related angles

Key point · Chapter path. **Objectives.** Convert from degrees to radians; place points; calculate standard sines and cosines; use the symmetries of the circle.

Prerequisites. Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Yearly reference point. 2–11 March: 8 hours

Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Perimeter of a circle of radius 1?
2. How many degrees is a quarter turn?
3. Calculate $\sqrt{\frac{1}{2}}$.
4. What are the x- and y-coordinates of the reflection of (a, b) in the x-axis?
5. Simplify $\frac{5\pi}{3} - 2\pi$.
6. In a right triangle, opposite side 3 and hypotenuse 5: sine?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present an equality of related angles, naming the symmetry that justifies it; explain what a radian measures.

Investigate before formalising

Construct, without a protractor, a point at angle $\frac{\pi}{3}$ then a point at angle $\frac{\pi}{4}$. Justify the exact values of their coordinates from the figures.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

10.1 Measuring an angle in radians

Key point · Definition and results. The unit circle is the circle with centre O and radius 1, oriented anticlockwise. One radian corresponds to an arc of length 1 on this circle. A full turn is 2π radians, a half turn π. On a circle of radius r, an arc length is $r\theta$ for a positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between central angle and arc length. The conversion is $180^\circ = \pi$ radians; we multiply degrees by $\frac{\pi}{180}$.

Example 1. $60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2π cm.

Watch out. The length of an arc is not that of the chord. In trigonometric calculations, check the degree or radian mode of the tool.

Application. Convert 45°, 120°; give the arc for radius 2 m and angle $\frac{3\pi}{4}$.

10.2 Wrapping and the image of a real number

Key point · Definition and results. To each real number x corresponds a point M of the circle, obtained by an oriented path of length x starting from I(1;0). Two real numbers have the same image if their difference is an integer multiple of 2π. A negative number indicates a path in the clockwise direction.

Status and justification. Adding or removing a whole number of turns keeps the end point. Conversely, returning to the same point requires a whole number of turns.

Example 2. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: the point $(0, -1)$.

Watch out. Adding π gives the opposite point, not the same point.

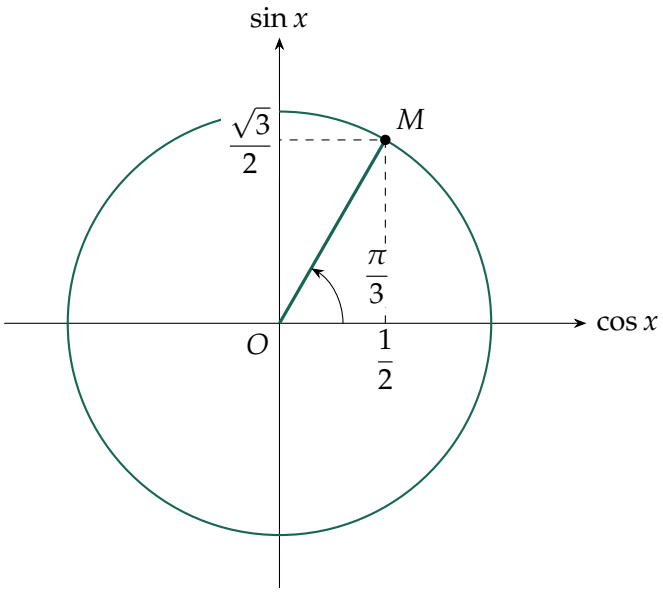
Application. Place the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to within one turn.

10.3 Sine, cosine and standard values

Key point · Definition and results. If M is the image of x, $M = (\cos x, \sin x)$. So $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$. For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are $1, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$ and the sines $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. At $\frac{\pi}{4}$, the two coordinates are equal and positive; Pythagoras gives $2a^2 = 1$, so $a = \frac{\sqrt{2}}{2}$. An equilateral triangle of side 1, cut by its altitude, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. The complementary angle gives $\frac{\pi}{6}$.

Example 3. The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}, \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors forming this angle equals its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.



For the angle $\frac{\pi}{3}$, the projections of M give its cosine and sine directly. The fractions are placed outside the lines of the drawing.

Watch out. The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

Application. If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

10.4 Related angles and symmetries

Key point · Definition and results. The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding $2k\pi$, k an integer, changes neither sine nor cosine.

Status and justification. Each equality can be read from the coordinates of the symmetric points: horizontal axis for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a catalogue without justification.

Example 4. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry from $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Watch out. Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

Application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

10.5 Approximating pi and putting it to use

Key point · Definition and results. A regular inscribed polygon with n sides in the unit circle has half-perimeter $n\sin(\frac{\pi}{n})$; a circumscribed polygon has half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

Status and justification. The inscribed perimeter is less than that of the circle and the circumscribed one greater: geometric bounds accepted. If c is the chord of one side, let $h = \sqrt{1 - \frac{c^2}{4}}$; the chord of the half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids a subtraction of nearby real numbers.

Example 5. For the inscribed hexagon, c=1 and half-perimeter 3. The circumscribed one gives $2\sqrt{3} \approx 3.4641$. By doubling the sides, we tighten the bounds. The supplied program starts from n=6 and c=1.

Watch out. Using a sine function directly with pi already known would check a calculation, but would not construct an approximation independent of pi.

Application. For n=6 and c=1, calculate the new chord after doubling.

Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 10.1 · Converting and measuring

We use the unit circle to measure angles in radians.

1. Convert 150° and -90° to radians, showing the conversion factor.
2. A circular sector has radius 4 cm and an angle of $\dfrac{\pi}{3}$ radians. Calculate the exact length of its arc.
3. Explain why an angle measure can be negative while a geometric arc length is positive.

Exercise 10.2 · Reducing angles

Consider the real numbers $\dfrac{17\pi}{6}$, $-\dfrac{7\pi}{4}$ and 5π.

1. For each, determine a representative in $[0, 2\pi)$ having the same image on the unit circle. Show the number of turns added or removed.
2. Place the three images on a circle and identify them by the starting real numbers.

Exercise 10.3 · Exact values

We look for the exact values of the cosine and sine of $\dfrac{2\pi}{3}$, $-\dfrac{\pi}{6}$ and $\dfrac{5\pi}{4}$.

1. Place the three images on the unit circle. For each, state the expected sign of its two coordinates.
2. Calculate the exact values using a standard angle and a symmetry. Present the answer in a table: angle, cosine, sine.

Exercise 10.4 · Recovering the sign

A real number x satisfies $\sin x = \dfrac{5}{13}$. Its image on the unit circle lies in the second quadrant, that is to the left of the y-axis and above the x-axis.

1. Determine the exact value of $\cos x$. Justify both its absolute value and its sign.
2. Explain why the value of $\sin x$ alone would not have been enough to choose the sign of $\cos x$.

Exercise 10.5 · A wheel

A wheel of radius 0.30 m rolls on horizontal ground without slipping. It turns through an angle of $\frac{5\pi}{3}$ radians. We look for the distance travelled by its axle in the direction of motion.

1. Calculate this distance in exact form, explaining the link with the arc length unrolled.
2. Explain why the length of a chord of the circle must not be used. Is the path of a point fixed on the rim of the wheel, seen from the ground, the same as that of the axle?

Exercise 10.6 · Chord and scalar product

On the unit circle with centre O, the point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.
2. Recover this length by applying the Al-Kashi relation to the triangle OAB. State the lengths and the angle used.

Exercise 10.7 · Reading an equation on the circle

We look for the real numbers x in the interval $[0, 2\pi)$ whose image on the unit circle has x-coordinate $\frac{1}{2}$.

1. Translate the condition on the x-coordinate into an equation involving $\cos x$.
2. Draw the corresponding vertical line on a circle, mark its intersections and give the two solutions. Check that they belong to the interval required.

Exercise 10.8 · Delayed review

Consider the angle of measure $\frac{25\pi}{4}$ radians. The answers must be exact, with no decimal approximation.

1. Reduce this angle, showing a whole number of complete turns.
2. Deduce its cosine and sine. Justify the values using the unit circle.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value supplied and the related angles, without a calculator.

CQFD, exercise 39 p. 220, questions 1, 2, 3 and 5. Do the four calculation and circle-reading questions; justify the answers on your paper.

MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $180°$ equals
   - A. 2π
   - B. π
   - C. $\frac{\pi}{2}$
   - D. 1
2. $\cos(\frac{\pi}{3}) =$
   - A. $\frac{1}{2}$
   - B. $\frac{\sqrt{3}}{2}$
   - C. 0
   - D. 1
3. $\sin(-\frac{\pi}{2}) =$
   - A. 1
   - B. 0
   - C. $\frac{1}{2}$
   - D. -1
4. $\cos^2 x + \sin^2 x =$
   - A. 0
   - B. 2
   - C. 1
   - D. x
5. The same point corresponds to angles differing by
   - A. π
   - B. $2k\pi$, k an integer
   - C. $\frac{\pi}{2}$
   - D. 1

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $\cos(\pi - x) =$
   - A. $\cos x$
   - B. $\sin x$
   - C. $-\cos x$
   - D. $-\sin x$
2. $\sin(\frac{\pi}{4}) =$
   - A. $\frac{\sqrt{2}}{2}$
   - B. $\frac{1}{2}$
   - C. $\frac{\sqrt{3}}{2}$
   - D. $\sqrt{2}$
3. Image of $\frac{3\pi}{2}$:
   - A. $(1,0)$
   - B. $(-1,0)$
   - C. $(0,1)$
   - D. $(0,-1)$
4. Arc of radius 3 and angle 2 radians: length
   - A. $\frac{2}{3}$
   - B. 6
   - C. $\frac{3}{2}$
   - D. 6π
5. Interior of the second quadrant: sign of the cosine
   - A. positive
   - B. zero everywhere
   - C. negative
   - D. cannot be determined

True or false? Justify.

1. $\cos(-x) = -\cos x$ for all x.
2. $\sin(x + 2\pi) = \sin x$.
3. The value of $\sin x$ always determines $\cos x$.

> **Key point · Self-assessment.** I reduce angles by complete turns; I read cosine and sine in the right order; I justify the signs; I keep the exact values.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

Solutions and teaching notes for chapter 10

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 2π. *Action:* Link radius, diameter and perimeter.
2. $90°$. *Action:* Divide a circle into four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Rationalise and check the square.
4. $(a, -b)$. *Action:* Draw the two points.
5. $-\dfrac{\pi}{3}$. *Action:* Put over a common denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Investigation activity

Equilateral triangle then isosceles right triangle. Pythagoras gives respectively height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Immediate applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ has the image of $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $\left(-\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2}\right)$ and $\left(\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right)$.
5. $h = \dfrac{\sqrt{3}}{2}$; new chord $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: method and results

Exercise 10.1 – Converting and measuring. $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates an orientation; a geometric length is positive.

Exercise 10.2 – Reducing angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2π.

Exercise 10.3 – Exact values. $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2}, -\frac{1}{2})$; $(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, in the form (cosine; sine).

Exercise 10.4 – Recovering the sign. $\cos^2 x = 1 - \frac{25}{169} = \frac{144}{169}$; cosine negative, so $-\frac{12}{13}$.

Exercise 10.5 – A wheel. Advance $r\theta = 0.3(\frac{5\pi}{3}) = \frac{\pi}{2}$ m. This corresponds to the unrolling of the arc; the chord joins two positions on the circle and does not have this length. A material point on the wheel follows a different path relative to the ground.

Exercise 10.6 – Chord and scalar product. $A(1,0)$, $B(-\frac{1}{2}, \frac{\sqrt{3}}{2})$; $AB^2 = \frac{9}{4} + \frac{3}{4} = 3$. Al-Kashi: $1 + 1 - 2\cos(\frac{2\pi}{3}) = 3$. So $AB = \sqrt{3}$.

Exercise 10.7 – Reading an equation on the circle. The vertical line with x-coordinate $\frac{1}{2}$ meets the circle at the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. The two points have opposite sines.

Exercise 10.8 – Delayed review. $\frac{25\pi}{4} = 6\pi + \frac{\pi}{4}$. Three complete turns do not change the image, so both values are $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a: $\cos(\frac{6\pi}{5}) = \cos(\pi + \frac{\pi}{5}) = -\frac{1+\sqrt{5}}{4}$. b: $\cos(-\frac{\pi}{5}) = \cos(\frac{\pi}{5}) = \frac{1+\sqrt{5}}{4}$, by evenness of the cosine.

CQFD, exercise 39 p. 220, questions 1, 2, 3 and 5. 1: $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$. 2: yes, $\cos(\frac{\pi}{3}) = \frac{1}{2} = 0.5$ is a decimal. 3: the principal measure in $(-\pi, \pi]$ is $-\frac{5\pi}{6}$. 5: $\sin(\frac{4\pi}{3}) = -\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half turn.
2. A. Equilateral triangle.
3. D. Lowest point of the circle.
4. C. Equation of the unit circle.
5. B. Complete turns.

MCQ 2.

1. C. Vertical symmetry.

2. A. Isosceles right triangle.
3. D. Three quarters of a turn.
4. B. $r\theta = 6$.
5. C. Negative x-coordinate inside the quadrant.

True or false

1. False: at 0, the two sides would be 1 and -1. The cosine is even.
2. True: a complete turn keeps the image.
3. False: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ have the same sine but opposite cosines.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.